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DEPARTMENT OF STATISTICS



M.Sc. QUANTITATIVE FINANCE
MSQF 534: CORPORATE INTERNSHIP AND VIVA

INTERNSHIP PROJECT REPORT ON

FINMETRICS SUITE

Automated Peer Benchmarking, Statistical Outlier Detection, and Multi-Ratio Financial Anomaly Engine for NSE Equities

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Project Repository:
`https://github.com/hi1r0/finmetrics-PU`

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1. Introduction and Organizational Context

1.1 Company Profile: IQ Business Advisors

The 6-week corporate internship was completed at **IQ Business Advisors**, a strategic financial advisory firm founded in 2024 by Mr. Naga Arjun and Mr. Rajesh Kanan. The organization specializes in three core operational domains:

- **Auditing & Process Optimization:** Streamlining internal controls, accounting systems, and statutory audits.
- **Strategic Financial Advisory:** Providing transaction diligence, portfolio valuation, and corporate restructuring.
- **Enterprise Analytical Automation:** Building custom quantitative workflows and decision-support algorithms.

Interns are embedded directly within live client delivery teams, translating corporate advisory needs into production software.

1.2 Project Overview and Core Objectives

FinMetrics Suite is an automated quantitative analysis platform developed to evaluate equities on the National Stock Exchange (NSE). Given any stock ticker, the engine retrieves filings, benchmarks the company against sector peers across 16 core financial ratios, and isolates univariate and joint structural anomalies using robust non-parametric statistics and machine learning.

The project fulfilled five core industrial and academic objectives:

1. **Automate Ingestion & Taxonomy Alignment:** Ingest exchange disclosures without manual spreadsheet extraction while harmonizing disparate accounting taxonomies across peer companies.
2. **High-Breakdown Outlier Detection:** Replace fragile Gaussian estimators with non-parametric methods (50% breakdown point) to insulate peer distributions against severe skewness and outlier masking.
3. **Median-Anchored Relative Benchmarking:** Normalize heterogeneous metric scales onto a unified 0–100 scale using sigmoid transformations anchored to sector medians and interquartile ranges.
4. **Multidimensional Risk Synthesis:** Formulate an explainable Composite Health Score and a 5-tier financial risk classification separating opportunity drivers from distress factors.
5. **Dual-Channel Enterprise Reporting:** Deploy a real-time Streamlit web dashboard alongside an automated, client-ready 6-tab Excel financial diagnostics workbook.

2. End-to-End System Architecture

FinMetrics Suite is architected as a modular four-stage analytical pipeline that decouples financial data ingestion, schema harmonization, dual statistical modeling engines, and multichannel reporting interfaces:

Pipeline Stage	`Core Module(s)`	Functional Responsibility
Stage 1: Data Ingestion	`ingestion_layer.py`	Queries exchange data for the target NSE ticker, dynamically extracts the sector peer universe, and ingests financial statements.
Stage 2: Schema Standardization	`preprocessing.py`	Resolves reporting discrepancies, aligns divergent accounting taxonomies across peers, and imputes or flags missing disclosures.
Stage 3: Dual Analytical Engines	`ratio_engine.py`	Computes 16 fundamental ratios across Profitability, Solvency, Growth, Liquidity, and Efficiency pillars.
	`outlier_engine.py`	Executes Tukey IQR fences, MAD-based Robust Z-Scores, and a 16-D Isolation Forest for univariate and joint anomaly detection.
	`benchmark_core.py`	Normalizes ratios via median-anchored sigmoids, dynamically redistributes missing weights, and computes the Composite Health Score.
Stage 4: Multichannel Output	`app.py, viz.py`	Serves an interactive Streamlit dashboard with Plotly distribution curves, radar charts, and peer percentile rankings.
	`report_generator.py`	Synthesizes analytics into an automated, formatted 6-tab Excel financial workbook for offline advisory audits.

3. Analytical and Statistical Methodology

3.1 Weighted Financial Ratio Engine (`ratio_engine.py`)

To evaluate corporate stability comprehensively, FinMetrics Suite implements 16 standardized fundamental metrics structured across five domain-weighted fundamental pillars:

Fundamental Pillar	Base Weight	Component Financial Ratios
Profitability	30%	Gross Profit Margin, Operating Margin, Net Profit Margin, Return on Equity (ROE), Return on Assets (ROA)
Solvency	20%	Debt-to-Equity Ratio (D/E), Total Debt Ratio (D/A), Interest Coverage Ratio (EBIT/Interest)
Growth	20%	YoY Revenue Growth, Net Income Growth, Operating Cash Flow (OCF) Growth
Liquidity	15%	Current Ratio (CA/CL), Quick Ratio ((Cash + Rec)/CL), Cash Ratio (Cash/CL)
Efficiency	15%	Total Asset Turnover (Rev/Assets), Inventory Turnover (COGS/Inv), Receivables Turnover

3.2 Non-Parametric Univariate Outlier Detection (`outlier_engine.py`)

Financial distributions in equity markets exhibit significant skewness and heavy fat tails. Classical parametric estimators (sample mean $\bar{x}$ and standard deviation s) possess a **0% breakdown point**—a single corrupted or extreme corporate disclosure can artificially inflate sample variance, producing *outlier masking* where moderate anomalies remain undetected. FinMetrics Suite utilizes two non-parametric estimators possessing a **50% breakdown point**:

- **Tukey Interquartile Range (IQR) Outer Bounds:** Non-outlier boundaries are derived from the 25th percentile (Q_1) and 75th percentile (Q_3):

$$\text{IQR} = Q_3 - Q_1, \qquad \text{Fences} = [Q_1 - 1.5\ \text{IQR},\ \ Q_3 + 1.5\ \text{IQR}] \tag{1}$$

- **Robust Z-Score via Median Absolute Deviation (MAD):** Deviations are scaled relative to the sector median $\tilde{x} = \text{median}(X)$ using the Gaussian consistency factor $0.6745 \approx 1/\Phi^{-1}(0.75)$:

$$\text{MAD} = \text{median}\left(|X_i - \text{median}(X)|\right), \qquad Z_{\text{robust}} = 0.6745 \times \frac{X_i - \text{median}(X)}{\text{MAD}} \tag{2}$$

Univariate Decision Criterion

A ratio is flagged as an individual statistical outlier if $|Z_{\text{robust}}| > 3.5$ or if it lies strictly outside the Tukey outer fences $[Q_1 - 1.5\ \text{IQR}, Q_3 + 1.5\ \text{IQR}]$, preventing extreme market distortions from biasing benchmark baselines.

3.3 Multivariate Anomaly Detection via Isolation Forest

While univariate detectors identify isolated metric spikes, corporate stress frequently manifests as structural discrepancies across multiple ratios (e.g., accelerating revenue growth coupled with deteriorating operating cash flow and surging leverage). To uncover hidden joint anomalies, `outlier_engine.py` applies an unsupervised **Isolation Forest** across the 16-dimensional ratio space:

- **Recursive Hyperplane Partitioning:** An ensemble of T isolation trees recursively partitions peer points using random axis-aligned cuts. Anomalous points occupy sparse regions and are isolated at significantly shallower tree depths $h(x)$.
- **Normalized Anomaly Formulation:** Using the average path length of unsuccessful searches in a Binary Search Tree $c(n) = 2\left(\ln(n-1) + 0.5772156649\right) - \frac{2(n-1)}{n}$ for peer size n, the ensemble anomaly score $s(x, n)$ is:

$$s(x, n) = 2^{-\frac{\mathbb{E}[h(x)]}{c(n)}} \tag{3}$$

Multivariate Anomaly Decision Threshold

A firm is classified as a multi-ratio structural anomaly when $s(x, n) \geq 0.60$. If $s(x, n) < 0.50$, the firm's overall financial posture is confirmed to be structurally concordant with its sector peer universe.

4. Peer Benchmarking, Risk Classification, and Reporting

4.1 Sigmoid Normalization and Dynamic Weight Redistribution (`benchmark_core.py`)

Financial ratios exhibit vastly different scales (margins as percentages, turnovers as multipliers, debt as leverage ratios). To map heterogeneous metrics onto an equitable 0–100 scale without distortion from extreme peer outliers, raw ratios are transformed using a tuned sigmoid function anchored directly to the sector median and IQR:

$$S(x) = \frac{100}{1 + \exp\left(-k \, \frac{x - \text{median}(X)}{\text{IQR}}\right)} \tag{4}$$

where k governs sensitivity around the sector median. For leverage and debt metrics where higher values indicate elevated financial vulnerability, the **Inverted Metric Transformation** is enforced:

$$S_{\text{inv}}(x) = 100 - S(x) \tag{5}$$

When specific accounting line items are unavailable or structurally inapplicable for a sector (e.g., inventory turnover in technology services), **Dynamic Weight Redistribution** re-allocates missing weights proportionally across all available metrics within that pillar:

$$w_i' = \frac{w_i}{\sum_{j \in \mathcal{A}} w_j}, \qquad \text{where } \mathcal{A} = \{j : \text{Ratio } j \text{ is reported}\} \tag{6}$$

This mathematical guarantee preserves total pillar integrity without falsely penalizing non-reporting firms.

4.2 Composite Health Score, 5-Tier Risk Classification & Explainable Diagnostics

Normalized metrics are aggregated across the five domain pillars into a **Composite Health Score (CHS)** ($0 \leq \text{CHS} \leq 100$) and mapped onto a rigorous 5-Tier Risk Framework:

- **Tier 1 (Exceptional, CHS ≥ 80):** Industry-leading margins, pristine solvency buffers, and elite operating cash flows.
- **Tier 2 (Strong, $65 \leq$ CHS < 80):** Robust peer standing with healthy fundamentals across majority pillars.
- **Tier 3 (Moderate, $50 \leq$ CHS < 65):** Balanced median-tracking profile with selective operational trade-offs.
- **Tier 4 (Vulnerable, $35 \leq$ CHS < 50):** Sub-par liquidity or deteriorating margins requiring credit surveillance.
- **Tier 5 (Distress, CHS < 35):** Severe capital depletion, structural leverage anomalies, or critical insolvency flags.

Rather than an opaque rating, the system generates **Explainable Attribution Diagnostics** that decompose total score deltas relative to the sector median into **Opportunity (+)**, **Neutral (∼)**, and **Risk (−)** drivers (e.g., *Liquidity Outperformance: +4.2 pts; Operating Margin Drag: −8.5 pts*).

4.3 Multichannel Output Interfaces

FinMetrics Suite provides two complementary enterprise interfaces:

1. **Interactive Web Dashboard (`app.py`, `viz.py`):** A Streamlit and Plotly application providing real-time ticker lookup, interactive sector peer distribution curves, pillar radar charts, and anomaly inspection tables.
2. **Automated 6-Tab Excel Diagnostics Workbook (`report_generator.py`):** Generates a client-ready multi-tab financial audit report comprising: *1. Executive Summary*, *2. Raw Ratio Matrix*, *3. Sigmoid Normalized Scores*, *4. Peer Percentile Rankings*, *5. Anomaly Audit Log*, and *6. Pillar Weight Distribution*.

5. Conclusion and Professional Reflections

FinMetrics Suite successfully bridges rigorous statistical theory, machine learning, and enterprise financial software engineering. By replacing fragile Gaussian assumptions with high-breakdown non-parametric estimators (Tukey IQR, MAD-based Robust Z-Scores) and unsupervised Isolation Forests, the platform insulates peer benchmarking against market skewness and outlier masking. Median-anchored sigmoid normalization and dynamic weight redistribution further guarantee fair, transparent, and explainable credit and equity risk assessments.

The 6-week corporate internship at IQ Business Advisors provided hands-on experience in productionizing quantitative models for real-world client workflows. The project achieved all core learning objectives of MSQF 534 (CO1–CO5), demonstrating the fusion of academic financial mathematics with automated corporate advisory solutions.